Adaptive Step Size Strategies for Inexact First-Order Methods under Strong Convexity

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Abstract

First-order optimization methods are widely used in large-scale optimization and machine learning due to their simplicity and scalability. However, in many practical settings, exact gradient information is unavailable, and algorithms must operate with inexact or approximate gradients arising from numerical errors, truncation, or partial computations. Understanding how such inexactness affects convergence behavior is therefore essential for designing reliable and efficient optimization algorithms.

This project systematically analyzes the convergence behavior of inexact first-order methods under strong convexity, with particular emphasis on how error structures and step size strategies jointly determine convergence speed and final solution accuracy. We study inexact gradient descent using a deterministic error model and derive a unified recursive inequality that characterizes the interplay between descent and gradient error. Based on observable gradient variability, we propose and compare two adaptive step size strategies: a continuous scaling rule based on gradient discrepancy, and a threshold-based switching rule that dynamically alternates between constant and diminishing step sizes.

We show that with constant step sizes, bounded gradient errors lead to convergence to a neighborhood of the optimum, yielding a quantifiable error floor, whereas diminishing step sizes eliminate this error floor and ensure convergence to the exact minimizer. The proposed adaptive strategies successfully interpolate between these behaviors, maintaining fast initial progress while preserving asymptotic convergence when gradient errors persist. Overall, this study demonstrates that the adverse effects of gradient inexactness in first-order methods are not inherent, but depend critically on step size design, providing both theoretical insight and practical guidance for error-tolerant optimization.

1 Introduction

1.1 Problem Statement and Context

First-order optimization methods, such as gradient descent and its variants, form the backbone of modern large-scale optimization and machine learning.

Their popularity stems from their simplicity, low per-iteration cost, and scalability to high-dimensional problems. In idealized theoretical settings, these methods are typically analyzed under the assumption that exact gradient information is available at every iteration.

In practice, however, this assumption is often violated. Gradients may be computed approximately due to numerical truncation, early stopping in inner solvers, finite-precision arithmetic, subsampling, or distributed and asynchronous computation. As a result, optimization algorithms frequently operate with inexact gradient information, which can significantly affect their convergence behavior.

Understanding how gradient inexactness impacts first-order methods is therefore a fundamental and practically relevant problem. In particular, it is crucial to determine under what conditions convergence can still be guaranteed, how fast convergence occurs, and whether inexactness leads to an unavoidable loss in solution accuracy. These questions are central to the design of reliable and efficient optimization algorithms in realistic computational environments.

1.2 Background and Theoretical Framework

This study focuses on smooth, strongly convex optimization problems, a class that admits a unique global minimizer and enjoys well-understood convergence properties under exact first-order methods. Strong convexity ensures a quadratic growth condition around the optimum, while smoothness provides control over the variation of gradients.

Under these assumptions, classical gradient descent with a constant step size achieves linear convergence to the optimal solution when exact gradients are available. The analysis typically relies on descent lemmas and Lyapunov-type inequalities that relate function value decrease to gradient norms. When gradients are inexact, the update direction can be modeled as the sum of the true gradient and an error term. This perspective naturally leads to a modified recursion in which the descent effect of the gradient competes with the accumulation of gradient errors.

Step size selection plays a critical role in this balance, as it simultaneously controls the strength of descent and the magnitude of error amplification. This theoretical framework provides a natural lens through which the robustness of first-order methods to gradient inexactness can be analyzed.

1.3 Review of Existing Literature

The effect of gradient errors has been studied across several research communities. Early work in stochastic approximation, beginning with Robbins and Monro, established convergence guarantees under diminishing step sizes and unbiased noise assumptions. Subsequent developments extended these ideas to stochastic gradient methods widely used in machine learning.

In deterministic optimization, gradient inexactness has been explored in the context of inexact Newton and proximal methods, where controlled accuracy requirements are imposed on inner solvers. More recently, the concept of inexact oracles has been formalized to analyze first-order methods under bounded or structured inaccuracies. These studies have shown that persistent gradient errors may prevent convergence to the exact optimum when constant step sizes are used, often resulting in convergence to a neighborhood of the solution. In contrast, diminishing step sizes can restore convergence, albeit at the expense of slower progress.

Despite these advances, existing analyses often focus on either asymptotic guarantees or worst-case complexity bounds. The interaction between error structure, step size strategy, and practical convergence behavior is not always made explicit, and adaptive mechanisms based on observable information are less commonly explored in a unified and transparent manner.

1.4 Research Gap and Rationale

While it is well understood that gradient inexactness can lead to an error floor under constant step sizes, and that diminishing step sizes can mitigate this effect, there remains a gap in understanding how to balance these two regimes in practice. Classical diminishing step size rules are often overly conservative, leading to slow convergence even when gradient errors are small or transient.

Moreover, many theoretical results assume prior knowledge of error magnitudes or rely on stochastic assumptions that may not hold in deterministic or numerically induced error settings. There is therefore a need for principled step size strategies that can adapt to the observed behavior of inexact gradients without requiring explicit access to the underlying error. Addressing this gap is important both theoretically and practically, as it clarifies the robustness limits of first-order methods and provides guidance for designing algorithms that are both efficient and error-tolerant.

1.5 Research Objectives and Hypotheses

The objectives of this study are threefold. First, we aim to provide a unified convergence analysis of inexact gradient descent for smooth, strongly convex functions, explicitly characterizing how gradient errors and step sizes jointly influence convergence rates and final accuracy. Second, we seek to distinguish between different error regimes, such as summable and bounded persistent errors, and to quantify their effects on convergence behavior, including the emergence of error floors. Third, we propose and investigate adaptive step size strategies based on observable gradient variability.

By comparing continuous scaling rules and threshold-based switching rules, we examine whether adaptive step sizes can preserve fast initial convergence while ensuring asymptotic robustness to gradient inexactness. We hypothesize that step size adaptation guided by observable gradient instability can interpolate between constant and diminishing step size behaviors, thereby mitigating error-induced stagnation without sacrificing efficiency. Through both theoretical analysis and numerical experiments, this study aims to validate this hypothesis and contribute to the broader understanding of error-aware first-order optimization methods.

2 Methods

2.1 Research Design

This study adopts a theoretical and computational research design. Rather than relying on empirical datasets or human-subject experiments, we investigate the behavior of optimization algorithms through rigorous mathematical analysis and controlled numerical simulations.

The methodology consists of two complementary components: (i) theoretical convergence analysis of inexact first-order methods under strong convexity assumptions, and (ii) numerical experiments designed to validate theoretical predictions and to compare different step size strategies. This combined approach allows us to systematically examine the interaction between gradient inexactness and step size selection.

2.2 Materials and Data

No real-world datasets or human participants are involved in this study. All experiments are conducted on synthetic optimization problems that are fully specified and reproducible.

The primary test problems are smooth, strongly convex quadratic objectives of the form

$$f(x) = \frac{1}{2}x^\top Ax, \tag{1}$$

where $A \in \mathbb{R}^{n\times n}$ is a symmetric positive definite matrix. Such problems admit a unique global minimizer and allow precise control over the strong convexity parameter μ and the smoothness constant L.

Gradient inexactness is modeled deterministically by injecting bounded error terms into the gradient evaluations. All numerical experiments are implemented in Python using standard scientific computing libraries.

2.3 Inexact Gradient Descent Procedure

We consider the unconstrained optimization problem

$$\min_{x\in\mathbb{R}^n} f(x), \tag{2}$$

where f is assumed to be μ-strongly convex and L-smooth. Let x^* denote the unique minimizer of f.

Instead of accessing the exact gradient $\nabla f(x_k)$ at iteration k, we assume an inexact first-order oracle of the form

$$g_k = \nabla f(x_k) + e_k, \tag{3}$$

where $e_k \in \mathbb{R}^n$ represents a deterministic gradient error.

The inexact gradient descent update is given by

$$x_{k+1} = x_k - \alpha_k g_k, \tag{4}$$

where $\alpha_k > 0$ denotes the step size at iteration k. The algorithm is initialized at a point x_0 away from the minimizer to ensure nontrivial convergence behavior.

2.4 Adaptive Step Size Strategies Based on Gradient Stability

The theoretical analysis shows that step size selection plays a critical role in balancing descent progress and the amplification of gradient errors. While constant step sizes may lead to an error floor and diminishing step sizes can be overly conservative, practical optimization problems often exhibit unknown and time-varying levels of gradient inexactness. This motivates the use of adaptive step size strategies based on observable gradient behavior.

Since the true gradient error e_k is not directly observable, we rely on quantities that can be computed from available information. In particular, we use the variation of consecutive inexact gradients as a proxy for gradient stability.

We define the gradient discrepancy

$$d_k := \|g_k - g_{k-1}\|, \tag{5}$$

and the relative gradient instability

$$r_k := \frac{\|g_k - g_{k-1}\|}{\|g_k\| + \varepsilon_0}, \tag{6}$$

where $\varepsilon_0 > 0$ is a small constant introduced to avoid numerical instability.

Based on these observable measures, we propose and study two adaptive step size rules.

Method I: Continuous Scaling Rule. The first strategy adjusts the step size smoothly according to the magnitude of the gradient discrepancy:

$$\alpha_k = \text{clip}\left(\frac{c}{1 + d_k},\ \alpha_{\min},\ \alpha_{\max}\right), \tag{7}$$

where $c > 0$ is a scaling parameter, $\alpha_{\min} > 0$ is a minimum step size, and $\alpha_{\max} \leq 1/L$ ensures numerical stability. This rule decreases the step size when gradient variability is large and allows larger steps when the gradient becomes more stable.

Method II: Threshold-Based Switching Rule. The second strategy employs a threshold-based mechanism that switches between constant and diminishing step sizes:

$$\alpha_k = \begin{cases} \alpha_{\max}, & r_k \le \tau, \\ \min\left\{\alpha_{\max}, \dfrac{c}{k+1}\right\}, & r_k > \tau, \end{cases} \tag{8}$$

where $\tau \in (0, 1)$ is a prescribed threshold. When the observed gradient instability is small, a constant step size is used to promote fast convergence. When instability exceeds the threshold, the method switches to a diminishing step size to suppress the accumulation of gradient errors.

Both adaptive strategies ensure that the step size remains bounded and rely solely on observable quantities. Their convergence behavior and empirical performance are evaluated in the subsequent results section.

2.5 Data Analysis and Evaluation Metrics

Algorithm performance is evaluated using quantitative convergence metrics. The primary measures include:

- the suboptimality $f(x_k) - f^*$,
- the distance to the minimizer $\|x_k - x^*\|$,
- the evolution of the step size α_k.

All quantities are analyzed across iterations and visualized on logarithmic scales where appropriate to highlight convergence rates and steady-state behavior. Comparisons are made between constant, diminishing, and adaptive step size strategies to assess convergence speed, robustness to gradient errors, and the presence or elimination of error floors.

3 Results

This section presents the main findings of the study. We first report theoretical results characterizing the convergence behavior of inexact gradient descent under different step size strategies. We then present numerical results that empirically validate the theoretical analysis and compare fixed and adaptive step size methods.

3.1 Theoretical Results

We consider the inexact gradient descent update

$$x_{k+1} = x_k - \alpha_k(\nabla f(x_k) + e_k),$$

where f is μ-strongly convex and L-smooth, and e_k denotes a deterministic gradient error. Let x be the unique minimizer of f, and define the suboptimality

$$\Delta_k := f(x_k) - f^*.$$

Core Descent Inequality. The following result establishes a fundamental recursion that governs the behavior of inexact first-order methods.
Theorem 3.1 (Core Descent Inequality). Let f be μ-strongly convex and L-smooth. If the step size satisfies $0 < \alpha_k \leq 1/L$, then

$$\Delta_{k+1} \leq (1 - \mu\alpha_k)\,\Delta_k + \frac{\alpha_k}{2}\,\|e_k\|^2.$$

This inequality highlights a trade-off between the contraction induced by strong convexity and the accumulation of gradient errors scaled by the step size. It serves as the basis for all subsequent convergence results.

Constant Step Size and Error Floor. We first examine the behavior under a constant step size.
Corollary 3.2 (Error Floor with Constant Step Size). Assume $\alpha_k \equiv \alpha \leq 1/L$ and that the gradient errors are uniformly bounded as $\|e_k\| \leq \varepsilon$. Then

$$\limsup_{k\to\infty} \Delta_k \leq \frac{\varepsilon^2}{2\mu}, \qquad \limsup_{k\to\infty} \|x_k - x^*\| \leq \frac{\varepsilon}{\mu}.$$

This result shows that persistent gradient errors prevent convergence to the exact minimizer when a constant step size is used, leading instead to convergence within a neighborhood whose size depends explicitly on the error magnitude.

Diminishing Step Sizes and Exact Convergence. We next consider diminishing step size sequences.
Theorem 3.3 (Convergence with Diminishing Step Sizes). Assume that $\|e_k\| \leq \varepsilon$ and that the step sizes satisfy

$$\sum_{k=0}^{\infty} \alpha_k = \infty, \qquad \sum_{k=0}^{\infty} \alpha_k^2 < \infty.$$

Then the iterates generated by inexact gradient descent satisfy

$$\lim_{k\to\infty} x_k = x^*, \qquad \lim_{k\to\infty} f(x_k) = f^*.$$

This theorem demonstrates that bounded but persistent gradient errors do not preclude exact convergence, provided that the step size decays sufficiently fast. The result indicates that the error floor observed with constant step sizes is not inherent to gradient inexactness, but rather a consequence of step size selection.

Implications for Adaptive Step Sizes. The above results suggest that step sizes that dynamically transition between constant and diminishing regimes may combine fast initial progress with asymptotic robustness. The adaptive step size strategies proposed in the Methods section are designed to exploit this observation by responding to observable gradient instability. Formal proofs of the above results are provided in Appendix A.

3.2 Numerical Results

We now present numerical experiments that validate the theoretical findings and illustrate the practical behavior of different step size strategies under inexact gradient information.

Experimental Setup. All experiments are conducted on strongly convex quadratic objectives of the form

$$f(x) = \frac{1}{2}x^\top Ax,$$

where A is a symmetric positive definite matrix. The spectrum of A is chosen to yield a nontrivial condition number, highlighting both transient and asymptotic behaviors. Gradient inexactness is modeled by injecting bounded, temporally correlated errors into the gradient evaluations, consistent with the inexact oracle assumptions used in the analysis. We compare the following step size strategies:

- a constant step size $\alpha = 1/L$,
- a diminishing step size $\alpha_k = c/(k + 1)$,
- an adaptive continuous scaling rule (Method I),
- an adaptive threshold-based switching rule (Method II).

Convergence Behavior. Figure 1 presents the evolution of the suboptimality $||f(x_k) - f^*||$ for the four step size strategies under correlated and bounded gradient errors.

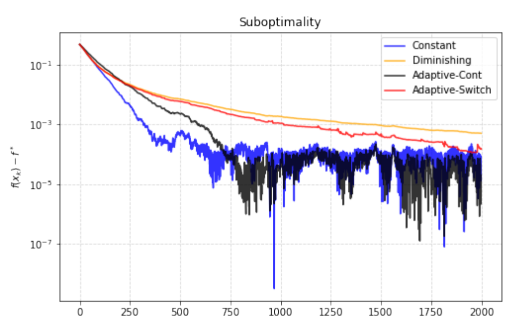

Figure 1: Suboptimality

All methods exhibit a comparable initial decrease, indicating that the descent direction is informative during the early phase of the optimization. However, pronounced differences emerge as the iterates approach the minimizer and gradient inaccuracies begin to dominate the dynamics. The constant step size method achieves the fastest initial reduction of the objective value. Nevertheless, its progress stagnates once the iterates enter a noise-dominated regime, resulting in persistent oscillations around a nonzero error floor. This behavior is consistent with classical analyses of inexact gradient descent, which predict convergence only to a neighborhood of the minimizer whose radius is proportional to the step size and the magnitude of the gradient error. In contrast, the diminishing step size strategy exhibits markedly smoother and more stable behavior. Although its convergence is slower throughout the optimization process, the progressive decay of the step size effectively suppresses the influence of persistent gradient errors. As a result, the diminishing rule continues to reduce the objective value with minimal oscillations, highlighting its robustness in the presence of correlated inexactness. The adaptive step size strategies are designed to interpolate between these two extremes. Both adaptive methods attain rapid initial descent comparable to the constant step size scheme by exploiting periods of apparent gradient stability. However, their long-term behavior differs significantly from that of the diminishing rule. Under correlated gradient errors, neither adaptive strategy consistently outperforms the constant or diminishing step sizes in the asymptotic regime. Instead, both remain sensitive to noise-induced fluctuations in successive gradient evaluations, which limits their ability to eliminate the error floor entirely.

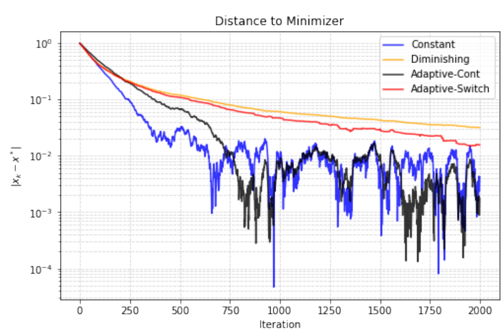

Figure 2: Distance to minimizer

Figure 2 reports the distance $||x_k - x^*||$ to the minimizer. The constant step size method converges to a fixed error neighborhood, while the diminishing rule steadily reduces this distance, albeit at a slower rate. The adaptive methods occasionally achieve closer proximity to the minimizer than the constant step size method; however, these improvements are transient and are not sustained uniformly across iterations.

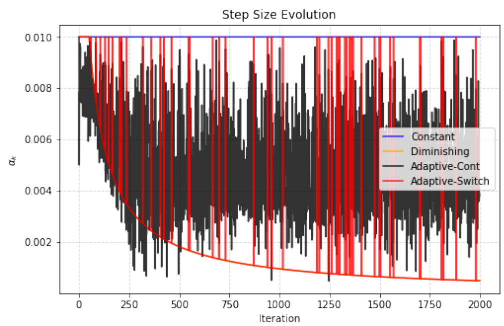

Figure 3: Step size evolution

Figure 3 illustrates the evolution of the step sizes. The continuous adaptive rule produces a highly reactive step size sequence, responding smoothly but sensitively to local variations in the approximate gradient. The threshold-based adaptive rule, on the other hand, exhibits pronounced switching behavior, alternating between aggressive updates and conservative steps. While both adaptive mechanisms respect classical stability limits, their dependence on noisy gradient differences ultimately constrains their asymptotic effectiveness.

Taken together, these results demonstrate that classical stability conditions alone are insufficient to ensure favorable asymptotic behavior when gradients are persistently inexact and temporally correlated. Adaptive step size strategies can significantly accelerate early progress and reduce the need for conservative parameter tuning, but diminishing step sizes remain the most reliable mechanism for suppressing long-term error accumulation.

3.3 Practical Scenarios Favoring Adaptive Step Sizes

The preceding experiments focus on the intrinsic convergence behavior of different step size strategies under correlated and persistent gradient errors. In this setting, diminishing step sizes exhibit superior asymptotic stability, while adaptive rules primarily improve transient performance.

We now turn to practical scenarios in which classical step size selection based on problem-dependent constants is either infeasible or overly conservative. In such cases, adaptive step size strategies can provide tangible benefits, despite their limitations in the asymptotic regime.

3.3.1 Mis-specified Lipschitz Constants

In practice, the Lipschitz constant of the gradient is often unknown and replaced by a conservative upper bound to ensure stability. While this guarantees convergence, it may result in step sizes that are far smaller than necessary and lead to slow progress in finite-horizon settings. Figure 4 illustrates a scenario in which the Lipschitz constant is severely overestimated. Both constant and diminishing step size rules, constructed using the incorrect bound, exhibit extremely slow

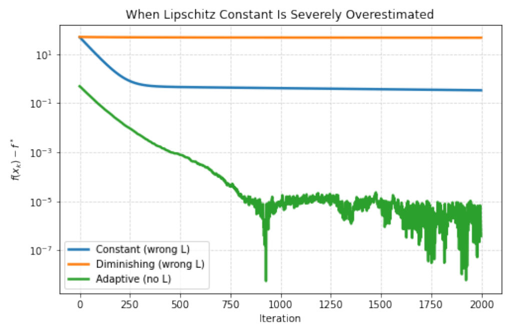

Figure 4: Overestimated Lipschitz constant

reduction of the objective value and fail to reach moderate accuracy within the allotted number of iterations. In contrast, the adaptive step size strategy does not rely on explicit knowledge of the Lipschitz constant and rapidly decreases the objective value by adjusting the step size based on observed gradient variations.

This example demonstrates that adaptive step size rules can effectively alleviate excessive conservatism caused by mis-specified problem constants, improving practical efficiency without altering the underlying optimization method.

3.3.2 Optimization with Finite-Difference Gradients

Finite-difference approximations provide a common means of computing gradients when analytic expressions are unavailable. Such approximations introduce deterministic and correlated errors that persist even near the minimizer, rendering classical step size selection based on exact smoothness assumptions overly conservative. Figure 5 compares different step size strategies when gradients are

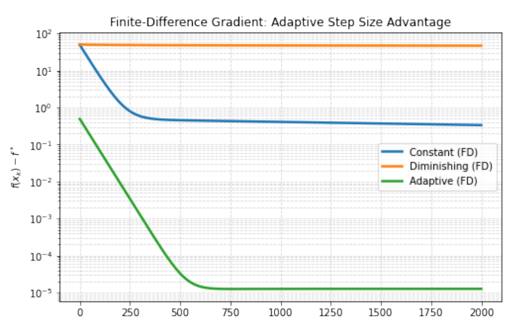

Figure 5: Finite-Difference Gradient

obtained via finite differences. Constant and diminishing step sizes exhibit slow progress due to the need for small step sizes to maintain stability. The adaptive strategy, however, exploits the relative stability of successive approximate gradients to automatically increase the step size during reliable phases and reduce it when approximation errors dominate.

These results indicate that adaptive step size strategies are particularly well-suited for optimization problems involving approximate gradients, where problem constants and error magnitudes are difficult to estimate accurately.

4 Discussion

4.1 Interpretation of Results

The experimental results provide clear evidence that the convergence behavior of inexact first-order methods is critically influenced by the choice of step size strategy. As predicted by the theoretical analysis, constant step sizes yield rapid initial progress but lead to convergence only to a neighborhood of the optimal solution when gradient errors persist. This phenomenon is consistently observed in both the suboptimality and distance metrics, manifesting as a nonzero error floor.

In contrast, diminishing step size rules eliminate the error floor and ensure asymptotic convergence to the exact minimizer. The numerical results, however, demonstrate that this robustness comes at the cost of slower convergence and conservative progress, particularly in the early iterations. This behavior reflects the classical trade-off between stability and efficiency in the presence of inexact gradient information.

The adaptive step size strategies investigated in this study aim to interpolate between these two regimes. By adjusting the step size based on observable gradient variability, the adaptive methods often achieve fast initial descent comparable to constant step sizes. However, under temporally correlated gradient errors, their long-term behavior remains sensitive to noise-induced fluctuations. While adaptive rules may transiently reduce error-induced stagnation, the asymptotic improvements are not uniformly sustained, and persistent oscillations can arise despite satisfaction of classical stability conditions.

Overall, the results indicate that adaptivity alone is insufficient to guarantee superior asymptotic performance in the presence of inexact gradients, highlighting the importance of robustness to noise structure rather than step size aggressiveness alone.

4.2 Comparison to Existing Literature

The observed behavior is consistent with classical results in stochastic approximation and inexact optimization. Foundational work by Robbins and Monro, as well as subsequent analyses of stochastic gradient methods, establishes that diminishing step sizes are essential for exact convergence under persistent noise. Similarly, studies on inexact oracles and deterministic gradient errors demonstrate that constant step sizes generally lead to convergence to a bounded error neighborhood.

The present study complements these results by providing a unified and transparent comparison between constant, diminishing, and adaptive step size strategies under a deterministic and temporally correlated error model. Unlike many existing approaches that rely on stochastic assumptions or explicit error bounds, the adaptive rules considered here depend solely on quantities observable from successive gradient evaluations. This perspective aligns with recent interest in adaptive and error-aware optimization methods, while also revealing potential limitations of naive adaptivity in the presence of correlated errors.

4.3 Limitations

Several limitations of the current study should be acknowledged. First, both the theoretical analysis and numerical experiments are restricted to smooth, strongly convex objectives. While this setting allows for precise characterization of convergence behavior, the effectiveness of the proposed adaptive step size rules in nonconvex or nonsmooth problems remains an open question.

Second, the error model assumes bounded deterministic gradient inaccuracies. Although this model captures numerical errors and truncated computations, it does not explicitly incorporate stochastic noise or bias, which are prevalent in large-scale machine learning applications. Moreover, the adaptive strategies rely on heuristic parameters, such as thresholds and scaling constants, whose optimal selection may depend on problem structure and noise characteristics.

4.4 Implications

Despite these limitations, the results offer important insights for the design of practical first-order optimization algorithms. In particular, they demonstrate that error-induced stagnation is not an intrinsic limitation of inexact gradient methods, but rather a consequence of step size selection. At the same time, the experiments indicate that adaptive step size modulation must be designed with care, as sensitivity to gradient variability can amplify the effects of correlated errors.

These findings suggest that adaptive step size strategies should explicitly account for noise structure or incorporate smoothing and averaging mechanisms to achieve reliable asymptotic behavior. Diminishing step sizes, while conservative, remain a robust baseline when strong accuracy guarantees are required.

4.5 Future Research

Several directions for future research emerge from this study. An immediate extension is to investigate adaptive step size mechanisms that are explicitly aware of noise correlation or incorporate temporal filtering of gradient information. Another promising direction is to extend the analysis to stochastic gradient settings and to study the interaction between stochastic noise and deterministic inexactness.

From a theoretical standpoint, deriving sharper convergence bounds that quantify the trade-off between adaptivity and robustness would provide deeper insight into step size design. Finally, exploring connections between gradient stability measures and second-order information may lead to more refined adaptive schemes that bridge the gap between first-order and second-order optimization methods.

5 Conclusion

5.1 Summary of Key Findings

This study investigated the convergence behavior of inexact first-order optimization methods under strong convexity, with a particular focus on the interaction between gradient inaccuracies and step size selection. Both theoretical analysis and numerical experiments confirm that constant step size methods achieve rapid initial descent but converge only to a neighborhood of the optimal solution in the presence of persistent gradient errors. In contrast, diminishing step size rules guarantee exact convergence, albeit at the cost of slower progress and conservative behavior during early iterations.

To explore alternatives between these two extremes, we examined adaptive step size strategies based on observable gradient stability. The results indicate that such adaptive rules can improve transient performance and partially alleviate error-induced stagnation. However, under temporally correlated gradient errors, adaptivity alone does not uniformly eliminate asymptotic error or guarantee superior long-term behavior, highlighting the inherent limitations of naive adaptive step size heuristics.

5.2 Research Contribution

The main contribution of this work is a unified and transparent framework for analyzing inexact first-order methods through the lens of step size design. By deriving a simple recursive inequality that captures the interaction between descent dynamics and gradient error, we clarify the origin of error floors and the fundamental role of diminishing step sizes in suppressing long-term inaccuracy.

In addition, this study provides a systematic empirical comparison of constant, diminishing, and adaptive step size strategies under a deterministic and correlated error model. The proposed adaptive rules rely solely on observable quantities and require no prior knowledge of error magnitudes or stochastic assumptions, offering insight into the potential and the limitations of error-aware step size modulation.

5.3 Broader Implications

The findings of this study have broader implications for large-scale optimization and machine learning applications, where exact gradient information is often unavailable or expensive to obtain. The results suggest that robustness to gradient inexactness is not solely a property of the objective function or optimization algorithm, but is critically influenced by step size selection.

At the same time, the observed sensitivity of adaptive step size rules to noise structure underscores the need for principled design beyond classical stability conditions. Rather than serving as a direct replacement for diminishing step size schedules, adaptive strategies should be viewed as a foundation upon which more robust, noise-aware mechanisms can be developed.

5.4 Final Remark

In summary, this work demonstrates that gradient inexactness does not fundamentally preclude effective first-order optimization. Instead, it reveals that the balance between efficiency and robustness is governed by subtle interactions between step size dynamics and error structure. By making these interactions explicit, this study provides a foundation for the development of more reliable and adaptive optimization methods in the presence of imperfect gradient information.

6 Appendices

This appendix provides detailed theoretical proofs, algorithmic descriptions, and supplementary experimental information that support the main results of the paper. These materials are deferred here to improve the clarity and flow of the main text.

7 Proofs of Main Theoretical Results

7.1 Proof of the Core Descent Inequality (Theorem 3.1)

Let f be μ-strongly convex and L-smooth. By L-smoothness, for any $x, y \in \mathbb{R}^n$,

$$f(y) \le f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2. \tag{9}$$

Applying this inequality with $y = x_{k+1} = x_k - \alpha_k g_k$ and $g_k = \nabla f(x_k) + e_k$, we obtain

$$f(x_{k+1}) \le f(x_k) - \alpha_k \langle \nabla f(x_k), \nabla f(x_k) + e_k \rangle + \frac{L\alpha_k^2}{2}\|\nabla f(x_k) + e_k\|^2. \tag{10}$$

Expanding the squared norm yields

$$f(x_{k+1}) \leq f(x_k) - \alpha_k \|\nabla f(x_k)\|^2 - \alpha_k \langle \nabla f(x_k), e_k \rangle \\ + \frac{L\alpha_k^2}{2} \|\nabla f(x_k)\|^2 + L\alpha_k^2 \langle \nabla f(x_k), e_k \rangle + \frac{L\alpha_k^2}{2} \|e_k\|^2. \tag{11}$$

Assuming $\alpha_k \leq 1/L$, we have

$$-\alpha_k + \frac{L\alpha_k^2}{2} \leq -\frac{\alpha_k}{2}. \tag{12}$$

Applying Young's inequality to the cross terms gives

$$-\alpha_k \langle \nabla f(x_k), e_k \rangle + L\alpha_k^2 \langle \nabla f(x_k), e_k \rangle \leq \frac{\alpha_k}{4} \|\nabla f(x_k)\|^2 + \frac{\alpha_k}{2} \|e_k\|^2. \tag{13}$$

Combining the above inequalities yields

$$f(x_{k+1}) \leq f(x_k) - \frac{\alpha_k}{4} \|\nabla f(x_k)\|^2 + \frac{\alpha_k}{2} \|e_k\|^2. \tag{14}$$

By μ-strong convexity,

$$\|\nabla f(x_k)\|^2 \geq 2\mu(f(x_k) - f^{),(15)}$$

and therefore

$$f(x_{k+1}) - f^* \leq (1 - \mu\alpha_k)(f(x_k) - f^*) + \frac{\alpha_k}{2} \|e_k\|^2. \tag{16}$$

This completes the proof.

7.2 Error Floor under Constant Step Sizes (Corollary 3.2)

Assume $\alpha_k \equiv \alpha \leq 1/L$ and $\|e_k\| \leq \varepsilon$. From the recursion above,

$$\Delta_{k+1} \leq (1 - \mu\alpha)\Delta_k + \frac{\alpha}{2}\varepsilon^2. \tag{17}$$

Solving this linear recursion gives

$$\Delta_k \leq (1 - \mu\alpha)^k \Delta_0 + \frac{\varepsilon^2}{2\mu}\big(1 - (1 - \mu\alpha)^k\big). \tag{18}$$

Taking $k \to \infty$ yields

$$\limsup_{k \to \infty} \Delta_k \leq \frac{\varepsilon^2}{2\mu}, \tag{19}$$

establishing the existence of an error floor.

7.3 Convergence with Diminishing Step Sizes (Theorem 3.3)

Assume $\sum_{k=0}^{\infty} \alpha_k = \infty$ and $\sum_{k=0}^{\infty} \alpha_k^2 < \infty$, with $\|e_k\| \leq \varepsilon$. Then the error term in the recursion is summable, while the descent term persists. Standard stochastic approximation arguments imply $\Delta_k \to 0$, and hence $x_k \to x^*$.

8 Adaptive Step Size Algorithms

8.1 Continuous Scaling Rule (Method I)

Algorithm 1 Inexact Gradient Descent with Continuous Adaptive Step Size

Initialize x_0, g_{-1}
for $k = 0, 1, 2, \ldots$ **do**
  $g_k = \nabla f(x_k) + e_k$
  $d_k = \|g_k - g_{k-1}\|$
  $\alpha_k = \text{clip}\Big(\frac{c}{1+d_k}, \alpha_{\min}, \alpha_{\max}\Big)$
  $x_{k+1} = x_k - \alpha_k g_k$
end for

8.2 Threshold-Based Switching Rule (Method II)

Algorithm 2 Inexact Gradient Descent with Threshold-Based Adaptive Step Size

Initialize x_0, g_{-1}
for $k = 0, 1, 2, \ldots$ **do**
  $g_k = \nabla f(x_k) + e_k$
  $r_k = \|g_k - g_{k-1}\|/(\|g_k\| + \varepsilon_0)$
  if $r_k \leq \tau$ **then**
    $\alpha_k = \alpha_{\max}$
  else
    $\alpha_k = \min\{\alpha_{\max}, c/(k+1)\}$
  end if
  $x_{k+1} = x_k - \alpha_k g_k$
end for

9 Additional Experimental Details

All experiments are conducted on strongly convex quadratic objectives of the form $f(x) = \frac{1}{2}x^\top Ax$, where $A \in \mathbb{R}^{d\times d}$ is symmetric positive definite. Unless otherwise stated, we consider $d = 2$ to enable clear visualization of the optimization trajectories.

The eigenvalues of A are chosen to control the condition number $\kappa = L/\mu$.

Gradient errors are modeled as bounded, temporally correlated perturbations added to the exact gradient. Specifically, the error sequence $\{e_k\}$ satisfies $\|e_k\| \leq \varepsilon$ for all k, with correlation introduced via a first-order autoregressive process. This construction captures deterministic numerical inaccuracies and truncated computations while remaining consistent with the inexact oracle assumptions used in the analysis.

All methods are initialized from the same starting point and are run for a fixed number of iterations. Results are qualitatively robust with respect to initialization, step size parameters, and error magnitudes within a reasonable range.

References

[1] H. Robbins and S. Monro, A stochastic approximation method, *Annals of Mathematical Statistics*, 22(3):400–407, 1951.

[2] J. Nocedal and S. J. Wright, *Numerical Optimization*, Springer, 2nd edition, 2006.

[3] S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004.

[4] S. Bubeck, Convex Optimization: Algorithms and Complexity, *Foundations and Trends in Machine Learning*, 8(3–4):231–357, 2015.

[5] S. Han, Gradient Methods with Dynamic Inexact Oracles, arXiv:2007.01206, 2020.

[6] P. Vernimmen and F. Glineur, Convergence analysis of an inexact gradient method on smooth convex functions, In *Proceedings of ESANN*, 2024.

[7] R. Chen, X. Tang, and X. Li, Adaptive stochastic gradient descent method for convex and non-convex optimization, *Fractal and Fractional*, 6(12):709, 2022.

[8] Y. Malitsky and K. Mishchenko, Adaptive gradient descent without descent, *Mathematical Programming*, 2020.